

The Cost of Indecision:

Analyzing Hesitation in Pickleball Rallies



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Introduction

For our project, we decided to observe Pickleball; a fun and competitive game in which players stand on opposite sides of a net and use a paddle to hit a perforated plastic ball back and forth. Pickleball can be played with two players (single matches) or four players (double matches). A point starts with an underhand serve below the waist to the diagonal opponent's court. After the serve, each team must let the ball bounce once before continuing the rally. A point is awarded to whoever hits the ball last into their opponent's side. Games typically are played to eleven points.

Hesitation in pickleball refers to a visible moment of indecision in movement, positioning, or shot commitment and execution that delays a player's response and puts them at a tactical disadvantage in the rally. Pickleball is worth studying because hesitation may be an under-measured cause of point loss. Traditional pickleball statistics focus on the final outcome of a point; such as a winner or type of serve. But a missed shot can be due to a number of hesitations; such as indecision about moving forward, changing direction, or attacking on a short ball. Studying hesitation could provide a more detailed understanding of how indecision in movement, shot selection, or court positioning affects rally outcomes.

A majority of the previous literature on pickleball isn't measured quantitatively and mostly falls back onto traditional coaching methods. Prior research has shown that pickleball can be studied through point-to-point observational coding but the existing analysis tends to focus on more visible outcome variables. The article still leaves room for additional observational variables that explain how tactical breakdowns develop during a rally (Touhey et al., 2024).

In the beginning, we hypothesized that the team or player with the most hesitations would lose the rally and we theorized that the type of hesitation influences rally results more than others. We go further into the hesitations in the next section.

Game ID	Rally ID	Serving Team	Receiving Team	Hesitation (S)	Hesitation (R)	Total Shots	Error Type	Rally Result
1	1	St. Louis Shock	Dallas Flash	yes	yes	9	positional	0
1	2	Dallas Flash	St. Louis Shock	yes	no	12	positional	0
1	3	Dallas Flash	St. Louis Shock	yes	yes	10	execution	0
1	4	St. Louis Shock	Dallas Flash	no	no	10	positional	0
1	5	St. Louis Shock	Dallas Flash	no	no	4	execution	1

For our observations, we primarily watched games hosted by Major League Pickleball. We found videos of their games on YouTube. We collected 200 observations in total (50 each) with 14 variables and each observation and row represents a doubles rally. As seen in the preview, we tracked games (Game ID), rally IDS (Rally ID), serving (Serving Team) and receiving teams (Receiving Team), hesitations (Hesitations) and how many (# of Hesitations, for both serving and receiving), total shots of serving and receiving teams (# Total of Shots, for serving and receiving), error type (Error Type), and rally result (Rally Result).

We assigned Game IDs based on which games were watched first by members and so on. We assigned Rally IDs based on consecutive order in their respective games. We recorded hesitations (Hesitations (S) for serving team, Hesitation (R) for receiving team) in each rally based on these parameters we created: player visibly pauses before moving, player stops in midcourt instead of continuing to kitchen after a playable shot, player retreats or drifts backwards under non-extreme pressure, player and/or partner show clear uncertainty over who should take over the ball. Hesitation was collected as either: yes if we observed hesitation, or no if we did not. Error type is another variable that we created, we categorized possible reasons as to why teams lose as mentioned earlier. Error type falls into three different categories: positional, decision, and execution. Positional losses occur due to player positioning or spacing on the court. Decision losses occur when a player makes a wrong tactical choice in their serve, such as a failed kitchen exchange or overaggression. Executional losses occur when a player fails to carry out a reasonable tactical choice, such as poor reset shot or missed volley. Rally result refers to whether or not the serving team won the point; 0 if they didn't, 1 if they did.

Each member had the same role of finding, reviewing, and recording their specific games. We all recorded our observations in a shared Google Sheets file in a 14x153 table.

Summary

Table 2:

Variable	Min	Max	Mean	SD
Total Shots	2	29	9.5	5.63
Shots by Server	1	15	4.97	2.88
Shots by Receiver	1	14	4.57	2.81
Hesitations (S)	0	3	0.56	0.7
Hesitations (R)	0	2	0.44	0.58
Hesitation Rate Per Shot	0	0.5	0.105	0.102

Table 2 is a summary of our key numeric rally-level variables. The average rally contained around 9-10 shots, with the serving team and the receiving team each contributing

around 5 shots on average, which indicates that most points involved several shots rather than one-shot errors. Hesitation events were relatively rare, with the serving team averaging 0.56 hesitations per rally, and the receiving team averaging 0.44 hesitations per rally. To account for the fact that a single hesitation is more impactful in a short rally than a long rally, we created a hesitation rate metric. The hesitation rate was defined for each rally as the total number of hesitations divided by the total number of shots, which allowed us to understand hesitations adjusted for the length of the rally. This metric is represented in the final row of Table 2, and the average hesitation rate for a rally is 0.105.

Table 3:

Variable - Value	Count	Percent
Hesitation (S) - No	110	55%
Hesitation (S) - Yes	90	45%
Hesitation (R) - No	120	60%
Hesitation (R) - Yes	80	40%
Rally Result - Loss	111	55.50%
Rally Result - Win	89	44.50%
Error Type - Decision	18	9%
Error Type - Execution	118	59%
Error Type - Position	64	32%

Table 3 is a summary of our key categorical rally-level variables. The serving team hesitated in 45% of rallies and the receiving team hesitated in 40% of rallies, so hesitations were present in some but not all points. The rally results column represents the serving team's wins and losses, so the serving team won 44.5% of rallies and therefore lost 55.5% of rallies. The majority of error types were executional (59%), followed by positional (32%), and finally decisional (9%).

Figure 1:

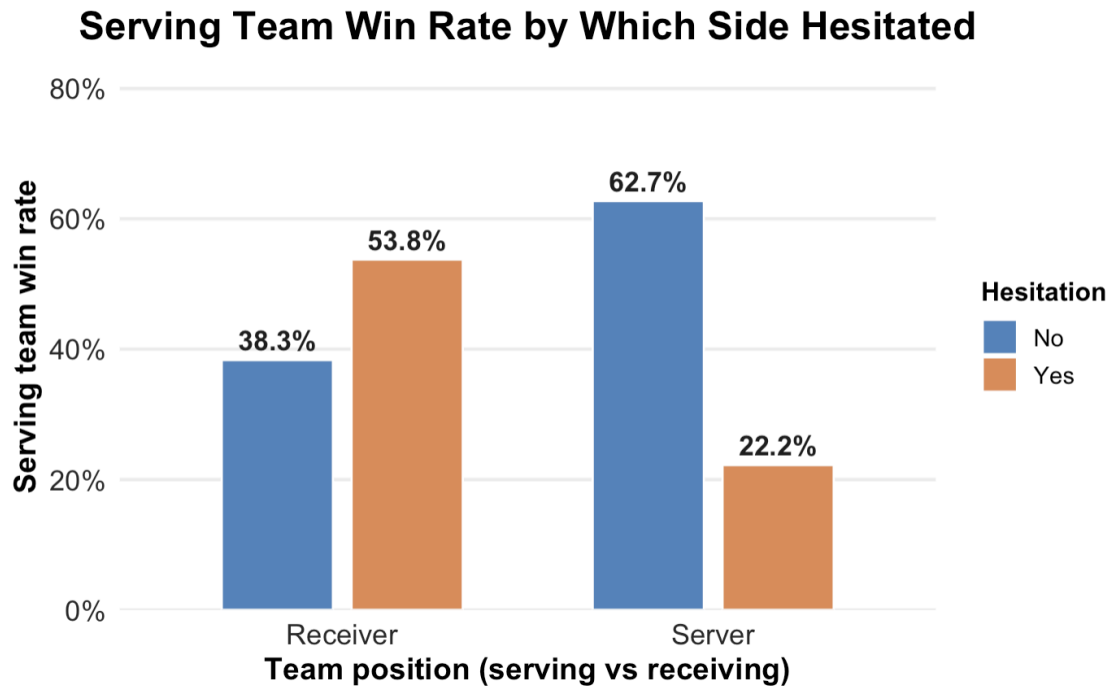


Figure 1 shows the serving teams win rate by team position, split by whether each side hesitated during the rally. When the server did not hesitate, they won 62.7% of rallies, but when they did hesitate, they only won 22.2% of rallies. In contrast, when the receiving team hesitated, the serving team's win rate was 53.8%, but when the receiving team did not hesitate, the serving team's win rate dropped to 38.3%. This indicates that hesitation is harmful for the team that hesitates and beneficial for the opponent.

Figure 2:

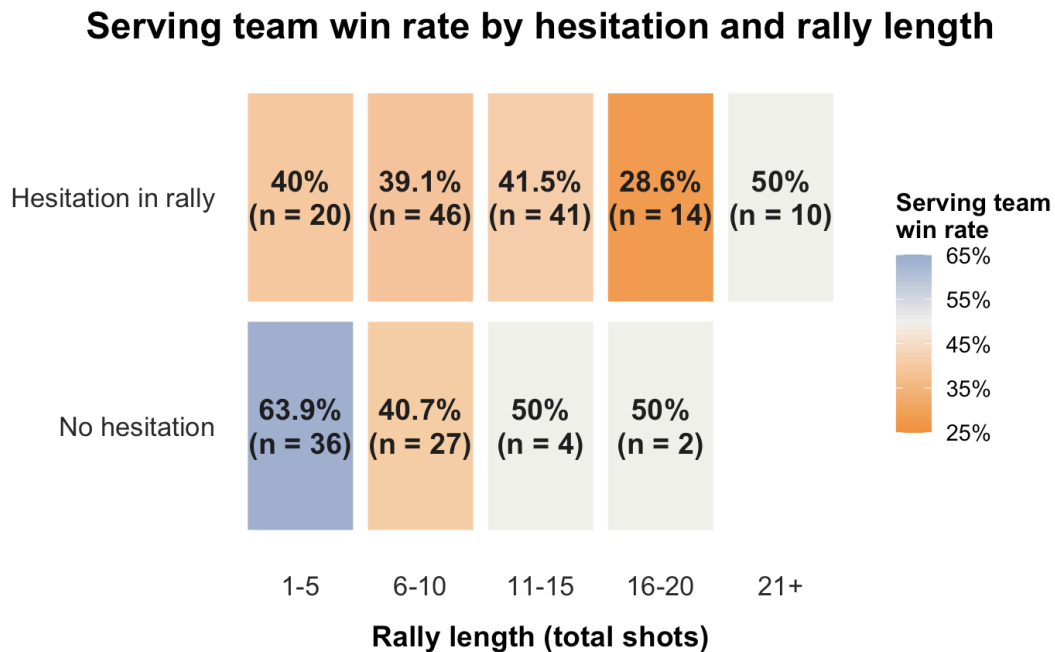


Figure 2 displays serving-team win rate across combinations of rally length and hesitation using a heatmap. Without hesitation, the serving team won 63.9% of short rallies, but as rally length increased, win rate decreased for the serving team into the 40% - 50% range. When the rally contained at least one hesitation from either team, win rates decreased for the serving team across all rally lengths. These patterns indicate that hesitations tend to occur in longer rallies and are associated with less favorable outcomes for the serving team.

Insights

Hesitation Impact

Based on Figure 1, it is clear that hesitation does have an affect on the serving team's win rate. More precisely, hesitation by the serving team is rare but extremely costly, while opponent hesitation is less rare and slightly beneficial for the serving team. Rather than looking at four separate percentages, we wanted a single metric that combines these effects and summarizes how favorable each rally was for the serving team. To do this, we designed a Hesitation Impact Score (HIS) that creates a penalty for server hesitation and a bonus for receiver hesitation so that each rally gets one number summarizing the impact of hesitation on the serving team.

For each rally, we let Y be 1 if the serving team won the rally and 0 if they lost, and we included our two indicator variables for hesitation: Server Hesitated and Receiver Hesitated. We then defined:

$$\text{HIS} = Y - \alpha(\text{Server Hesitated}) + \beta(\text{Receiver Hesitated})$$

Here α is a penalty that represents how much server hesitation reduces the serving team's win probability, while β is a bonus that represents how much receiver hesitation increases that probability. We used a logistic regression model predicting serving-team win probability with server and receiver hesitation as predictors to estimate α and β . Using this model we computed the predicted win probability when neither side hesitated, when one side hesitated, and when both sides hesitated. Then we let α be the decrease in predicted win probability when the server hesitated and β be the increase when the receiver hesitated. This method gave us values of $\alpha = 0.398$ and $\beta = 0.201$, so server hesitation reduces the predicted win probability by 39.8% while receiver hesitation increases it by 20.1%. The coefficient for server hesitation was statistically significant with a p-value of 1.5×10^{-8} , which indicates that server hesitation significantly reduces the log-odds of winning the rally even after controlling for opponent hesitation. The coefficient for receiver hesitation was also significant with a p-value of 0.00675, which indicates that receiver hesitation increases the log-odds for the serving team to win the point. With this definition, a higher HIS score corresponds to rallies where the server wins without hesitating or benefits from opponent hesitation, while a lower HIS score corresponds to rallies where the server hesitates and loses the point.

To analyze our data alongside our HIS metric, we split the data into 4 cases, and summarized the HIS scores per case below in Table 4.

Table 4:

Case	Rally Count	Mean HIS	SD HIS
Receiver Hesitation Only	41	0.98	0.42
No Hesitation on Either Side	69	0.54	0.50
Both Hesitation	39	0.08	0.46
Server Hesitation Only	51	-0.22	0.39

Based on the average HIS values, we see a clear ranking of rally types from most to least beneficial for the serving team. Rallies where only the receiver hesitated had the highest average HIS at 0.98, which indicates that these situations were beneficial for the server. Rallies with no hesitation from either side had a lower but still positive average HIS of 0.54, which reflects a slight advantage for the serving team. When both teams hesitated, the average HIS was 0.08, which suggests that hesitations from both sides cancel each other out and lead to no gain or loss for either side. Finally, rallies where only the server hesitated had an average HIS of -0.22, which aligns with our conclusion from Figure 1 that rallies where only the server hesitates are the most damaging to their ability to win.

Insight 2:

Error Type

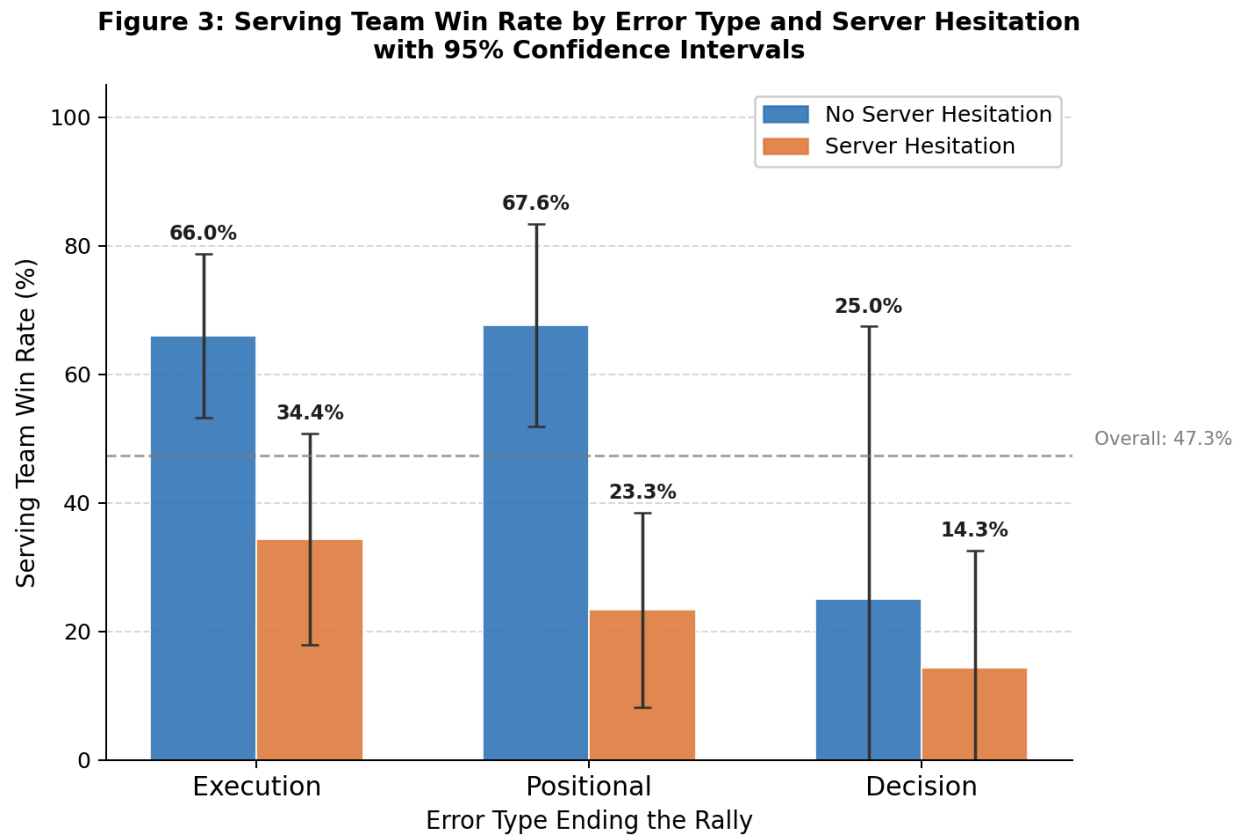
Our second insight examines whether the type of error that ends a rally (positional, executional, or decisional) is associated with different outcomes for the serving team, and whether hesitation by the server changes the kinds of errors that occur.

We first examined the serving team's win rate separately for each error type. When the rally ended in an execution error (a technically incorrect shot), the serving team won 54.1% of rallies ($n = 85$, 95% CI: [43.5%, 64.7%]). When it ended in a positional error (a player being out of position), the serving team won 46.9% of rallies ($n = 64$, 95% CI: [34.6%, 59.1%]). Most strikingly, when the rally ended in a decision error (a player choosing the wrong shot), the serving team won only 16.7% of rallies ($n = 18$, 95% CI: [-0.6%, 33.9%]). A chi-square test across all three groups confirms that these differences are statistically significant ($\chi^2 = 8.37$, $p = 0.015$), meaning error type and rally outcome are not independent of one another.

To identify which specific pairs are driving this result, we conducted pairwise two-proportion z-tests with Bonferroni correction (adjusted $\alpha = 0.0167$ for three comparisons). Decision errors produced significantly lower win rates than both execution errors (difference = -37.5 percentage points, $z = -3.63$, $p = 0.0003$, 95% CI: [-57.7%, -17.2%]) and positional errors (difference = -30.2 percentage points, $z = -2.80$, $p = 0.005$, 95% CI: [-51.3%, -9.1%]). By contrast, the difference between execution and positional errors was not statistically

significant ($p = 0.380$). This tells us that decision errors stand apart as uniquely costly since rallies ending in a decision error are strongly associated with the serving team losing the point.

Figure 3:



To understand why decision errors are so damaging, we examined rally length by error type. Rallies ending in a decision error were substantially longer on average (mean = 14.2 shots, SD = 4.88) compared to execution errors (mean = 7.4 shots, SD = 4.08) and positional errors (mean = 10.5 shots, SD = 6.20). A one-way ANOVA confirms that these differences in rally length are statistically significant ($F = 15.93$, $p < 0.0001$). This pattern suggests that decision errors tend to occur late in extended rallies, when players are under greater physical and mental pressure. A wrong shot choice at the end of a long rally is especially costly because it wastes the effort invested in the preceding exchange.

Finally, we found that server hesitation is not only associated with losing rallies, it also changes the type of error that ends those rallies. When the server did not hesitate, only 4.4% of rallies ended in a decision error, compared to 18.4% when the server did hesitate. A chi-square test confirms that the distribution of error types differs significantly depending on whether the server hesitated ($\chi^2 = 9.72$, $p = 0.008$). This is an important finding because it connects our two insights: hesitation does not just reduce the probability of winning, it specifically increases the

likelihood of a decision error, which is the most damaging error type of all. In other words, hesitation and poor decision-making appear to be part of the same underlying phenomenon: a breakdown in composure under pressure.

Taken together, these findings offer a concrete coaching implication. Since execution and positional errors do not significantly differ in their association with rally outcomes, training time should be disproportionately focused on reducing decision-making errors, particularly in long rallies where mental fatigue accumulates.

Limitations and Improvements

Looking back on our project, there are some ways we could have improved both our data collection and our analysis. First, hesitation events were relatively rare in our sample, which means that estimates of win probability and HIS in rallies are based on small counts and therefore have high uncertainty. Using a larger dataset or targeted sampling toward high-pressure situations with more hesitations would allow us to report more reliable estimates. Second, while we did have a clear definition of hesitation, different people might disagree on borderline cases. If we repeated this study, we would watch each other's matches to confirm that we all agree on the same label for every rally.

Beyond these limitations, our findings still provide meaningful evidence that hesitation plays a critical role in rally outcomes. Even with a relatively small sample, the consistent patterns observed across Figures 1 and 2 and the Hesitation Impact Score suggest that hesitation is not random noise, but a systematic factor influencing performance. In particular, the asymmetry between server and receiver hesitation highlights that hesitation is context dependent, with greater consequences when it occurs in more control-oriented positions such as serving. Additionally, our results linking hesitation to decision errors reinforce the idea that hesitation is not just a physical delay, but also reflects cognitive breakdowns under pressure. These patterns indicate that hesitation could serve as a useful intermediate variable for understanding how rallies unfold, bridging the gap between observable actions and final outcomes.

In the future, it would be beneficial to include variables like skill level, score at the time of the rally, or fatigue, as those could influence the likelihood of hesitation and the chance of winning the point and therefore could be potential confounding variables. It would also be beneficial to redo the study but at the shot level instead of the rally level as this would allow us to distinguish between early versus later hesitations and connect hesitations to shot types or locations on the court.

References

Touhey, D. C., Bozorgmehr, C. K., Tartibi, D. S., Smith, M. V., & Knapik, D. M. (2024). *Pickleball injuries in the aging athlete: A critical analysis review*. *Cureus*, 16(9), e69950.